

Sarthak Consul

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Experience

Matic Robots, Inc

Research Engineer, Mapping and Neural Networks

Mountain View, CA

April 2023 - Present

Training

- Fine-tuned and optimized large **vision-language (VLA)** and **vision-transformer (ViT)** foundation models, enabling the bot to locate and tidy up objects on the floor.
- Drove transition from CNN-based networks to **ViT**-based novel occupancy networks with **distillation**, enabling sub-voxel geometric and semantic predictions and eliminating irrecoverable failure modes.
- Achieved **23.7x training speedup** and **14x memory efficiency gain** by optimizing the multi-node distributed training pipeline with **DALI**, **DDP**, mixed-precision (**AMP**) training, and selective **torch.compile**.
- Improved ground truth and postprocessing algorithms for **8x** higher resolution and **30x** more robust maps.
- Designed a custom data format yielding **3x** faster dataloading, **[X]%** lower CPU overhead, and minimal egress, unlocking data streaming for cloud training.

Inference Optimization

- Spearheaded end-to-end inference optimization, leveraging **TensorRT** to compile PyTorch DAGs into **post-training quantized** engines for a **6.7x inference speedup**. Automated build and precision-tuning workflows (FP32 → FP16/INT8 kernel fusion), saving **20+ engineering hours/week** across the team.
- Engineered multi-model inference pipelines, **halving** latency through strategic pipelining, GPU memory pinning, and model preloading, delivering a **30%** improvement in bot responsiveness.
- Deployed hybrid CPU-GPU orchestration on edge with asynchronous host-device memory transfers, balancing device and host loads for **zero hang-time** and maximal throughput.
- Designed an efficient GPU workspace memory-sharing system, reducing engine runtime footprint by **400MB**.
- Authored custom **CUDA kernels** that outperformed TensorRT-optimized kernels by **18%** on the same workload.

Engineering

- Architected an end-to-end **data platform (Rust + Apache DataFusion)** over **[X]TB+** of multimodal robot logs – columnar Parquet storage with spatial/temporal indexing, async streaming dataloaders, and a UI for annotation, semantic search, dataset curation, and side-by-side model-output diffing. **Dual-mode** deployment (on-prem annotation cluster + cloud query instance) with pipelined I/O hiding fetch latency for sub-second response across **[X]M+** samples; adopted org-wide by **[X]+** engineers, cutting dataset-iteration cycles by **[X]x**.
- Led a team of 3 engineers to deploy data-parallel execution on NVIDIA **Jetson** SOMs, working across the stack – authored **Rust** TensorRT wrappers, CUDA stream managers, and efficient memory allocation schemes.
- Led development of post-processing and aggregation algorithms over NN predictions to build 2D/3D maps that are globally stable and locally dynamic (under **250ms latency**) with **1cm³ resolution** covering **12,000+ sq ft**, robust to varying lighting conditions, SLAM pose inaccuracies, and camera calibration noise.
- Single-handedly designed and implemented an automated Python→Ambarella vector-processor compiler supporting **130 operations**, significantly improving code readability and compilation reliability.

Meta

SWE Intern, Ad Product Optimization

Menlo Park, CA

Jun 2022 – Sept 2022

- Implemented a **group-DRO** loss for the production **ad-finder model's** online-learning loop, improving **distributional robustness under distribution shift** and reducing the worst-group traffic-partition performance gap by **15%** in a live A/B test.

IIT Bombay

Research Assistant

Mumbai, India

Jun 2020 – Aug 2021

- Designed a **single-round** group-testing algorithm for COVID-19 detection that reduced RT-qPCR tests needed by up to **10x**. Deployed across Indian testing facilities during the pandemic.

Technical Skills

Languages & Frameworks: Python, CUDA, PyTorch, Rust, C/C++, SQL, Bash, TensorFlow, Keras

ML Systems: Distributed training (DDP, AMP, DALI, **torch.compile**), post-training quantization (FP16/INT8), kernel authoring & fusion, NCCL, model distillation

Tools: Docker, Slurm, PyTorch Lightning, Git, NVIDIA (Nsight Systems/Compute, TensorRT, DALI, TAO)

Selected Publications

- [1] **Invalid Logic, Equivalent Gains: The Bizarreness of Reasoning in Language Model Prompting**
ICML Knowledge and Logical Reasoning Workshop, 2023.
- [2] **Compressed Sensing Approach to Group-testing for COVID-19 Detection.**
IEEE Open Journal of Signal Processing, 2021.
- [3] **Lower Bounds for Policy Iteration on Multi-action MDPs.**
59th IEEE Conference on Decision and Control, 2020.

Open Source

- **Streamable Gaussian Splat Viewer (Rust + WGPU/CUDA)** – High-performance **Rust** renderer for **streamable** 3D Gaussian-splat scenes: progressive LOD streaming over the network, on-the-fly decoding, and custom GPU rasterization kernels targeting **[X] FPS at 4K** on scenes with **[X]M+ splats** – **[X]x** faster and **[X]x** less memory than reference implementations. *WIP*.

Education

Stanford University

M.S. in Computer Science (AI Specialization)

Indian Institute of Technology Bombay

B.Tech (with Honors) in Electrical Engineering and Minor in CS